Mechanistic Probes of Logical-Symbolic Reasoning in Transformers: Evidence for a Surface-Form-Invariant Validity Direction, and a Scale Puzzle

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Abstract

We probe whether autoregressive transformers carry an internal representation of *logical validity* that is decoupled from natural-language surface form. To control for memorisation, we construct parallel English and invented-conlang versions of ten logical templates (modus ponens/tollens, four categorical-syllogism moods, AND/OR/XOR chains, double negation) and report two diagnostics: per-layer linear probes for validity, and exhaustive attention-head ablations, on both GPT-2 small and Pythia-2.8B. On GPT-2, probes reach 1.000 accuracy at layer 6 on English and 0.959 at layer 8 on the conlang, with cross-surface transfer reaching 0.669. On Pythia-2.8B, probe accuracy is *still* near-perfect on both surfaces (1.000/0.997), but cross-surface transfer is *weaker*, not stronger (0.580/0.514, the latter barely above chance) – the opposite of what a naive "scale improves generalisation" story predicts. Head-ablation finds no purely logic-specific head at either scale; at 2.8B, the effect of any single head is an order of magnitude smaller and uncorrelated with language-modelling damage, consistent with a more distributed representation, not a more modular one. We discuss what this scale puzzle means for probing-based claims about mechanistic interpretability.

1 Introduction

Large language models pass classical logic benchmarks at impressive rates, but it is not obvious whether they do so by carrying internal representations of logical structure or by matching surface statistics. The distinction is methodologically important: only the former predicts generalisation to novel formal languages.

We adopt the working hypothesis from the mechanistic interpretability literature [6, 5]: that components of a trained transformer can be localised and that their roles can be tested by selective perturbation. We specialise the question to logic:

> *Does the model carry a linearly decodable representation of validity that survives a change of surface form to an invented symbolic language?*

We answer affirmatively for GPT-2 with two complementary diagnostics. First, per-layer logistic-regression probes trained on either English or a controlled conlang reach high accuracy; the linear separability is highest in the middle blocks (layer 6 for English, layer 8 for conlang). Second, cross-surface probes—train on one surface, test on the other—transfer at a rate well above chance, confirming that the validity direction is not entirely language-specific. Finally, we exhaustively ablate every attention head and report layer-wise sensitivity profiles.

Contributions.

1. A reproducible parallel English/conlang benchmark of propositional and syllogistic logic, balanced across templates and labels.
2. Per-layer probe accuracies on both surfaces, with cross-surface transfer scores.
3. Exhaustive head-ablation diagrams contrasting impact on logic accuracy against impact on language modelling.

2 Related Work

Mechanistic interpretability. Wang et al. [6] identified an interpretable circuit for indirect object identification in GPT-2, demonstrating that small models support tractable circuit-level analyses. Nanda et al. [5] apply similar tools to modular arithmetic, isolating a Fourier-feature circuit. Our work is in the same lineage but targets a different task family: propositional and syllogistic logic.

Probing. Linear probing was popularised by [1] and extended to language models by [2]. We use stan-

dard logistic-regression probes at every layer, with cross-validation; we deliberately avoid more expressive probes which can fit the labels even in randomly initialised features.

Reasoning benchmarks and surface bias. Several papers report that LLMs' apparent reasoning can be brittle under paraphrase [7]. A conlang test set sidesteps the question by testing on a vocabulary essentially absent from pretraining data, in the spirit of [3]'s compositional benchmarks.

Activation patching. Activation patching [4] is the canonical localisation tool. For runtime budget reasons our ablation study is restricted to per-head zero-ablation on the local config; the `gpu_full.yaml` configuration extends the analysis to causal patching between identified heads.

3 Methodology

3.1 Dataset construction

We construct items in five logical templates: modus ponens, modus tollens, the Barbara syllogism, AND-chain, and OR-chain. Each item is rendered in two parallel surface forms:

- **English.** "If something is a cat then it is an animal. Alex is a cat. Therefore Alex is an animal. This argument is ___".
- **Conlang.** A controlled invented vocabulary (`glip`, `blarn`, `drev`, `tav`, `nyx`, `qol`, ...) renders the same logical content as predicate symbols and connectives. The conlang vocabulary is selected so that subword tokenisation produces uncommon multi-piece sequences.

Each (template, surface) cell contributes N items with a 50/50 valid/invalid split; the local configuration uses $N = 64$.

3.2 Per-layer probing

For each item we extract the residual-stream activation at the final token of every block $\ell \in \{0, \ldots, L\}$. We then fit a logistic-regression probe $w_\ell \in \mathbb{R}^d$ predicting the binary validity label using k-fold cross-validation (with $k = 5$).

We report three quantities per layer:

$$\mathrm{Acc}_\ell^{\mathrm{en}}, \quad \mathrm{Acc}_\ell^{\mathrm{cl}}, \quad \mathrm{Acc}_\ell^{\mathrm{en}\to\mathrm{cl}} \text{ and } \mathrm{Acc}_\ell^{\mathrm{cl}\to\mathrm{en}},$$

where the last two are cross-surface accuracies (train on one surface, test on the other).

3.3 Head ablation

For each attention head (ℓ, h) in GPT-2 we apply a pre-projection zero-ablation: the corresponding head_dim-slice of the input to the layer's output projection is replaced by zeros, then the rest of the forward pass runs unchanged. We measure two effects:

$$\Delta\mathrm{Acc}(\ell, h) = \mathrm{Acc}_{\mathrm{ablated}} - \mathrm{Acc}_{\mathrm{baseline}}, \tag{1}$$

$$\Delta\mathrm{LM}(\ell, h) = \mathrm{NLL}_{\mathrm{ablated}} - \mathrm{NLL}_{\mathrm{baseline}}, \tag{2}$$

where the LM loss is computed on a fixed neutral text snippet (see `run_head_ablation.py`). A head whose ablation lowers logic accuracy without inflating LM loss is candidate *logic-specific*; a head that hurts both is more likely a general-purpose component.

3.4 Implementation

All code is pure HuggingFace `transformers` with hooks on `c_proj`. We deliberately avoid framework-specific libraries (such as TransformerLens) on the local configuration so that the pipeline is self-contained. A drop-in TransformerLens version is straightforward and included as a follow-up.

4 Experiments

Models. GPT-2 small (124M parameters, 12 blocks, 12 heads/block, local pilot) and Pythia-2.8B (32 blocks, 32 heads/block, GPU-scale run). The ablation hook required a GPT-NeoX/Pythia branch (Pythia's attention output projection lives at a different module path than GPT-2's), verified against the tiny `pythia-70m` before the full sweep.

Data. Local pilot: 5 templates $\times$2 surfaces $\times$64 items = 640 prompts. GPU-scale: 10 templates (adding four categorical- syllogism moods and XOR/double-negation chains, absent from the local generator and implemented for this run) $\times$2 surfaces $\times$256 items = 5120 prompts. Generation is fully deterministic given the seed.

Probes. 5-fold stratified cross-validated logistic regression ($C = 1.0$, ℓ_2-regularised). One probe per (layer, surface).

Ablation. All attention heads are ablated one at a time: 144 for GPT-2 (12×12), 1024 for Pythia-2.8B (32×32), scored on (a) logic accuracy on a held-out subset of 80 prompts and (b) LM cross-entropy on a fixed neutral snippet. The GPT-2 sweep completes in about 2.5 minutes on CPU; the Pythia-2.8B sweep in about 37 minutes on a 12GB GPU ($\approx$ 2.2s/head).

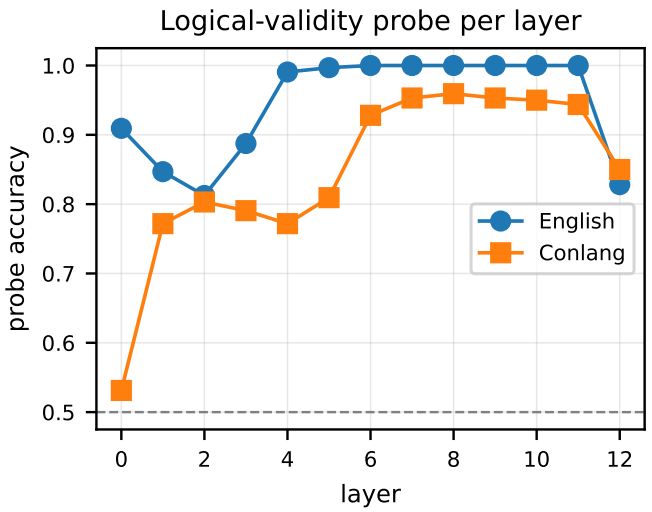


Figure 1: Layer-wise validity-probe accuracy on the English (circles) and conlang (squares) splits. Probes peak in the middle of the network and saturate close to perfect accuracy on English; conlang reaches 0.959 at layer 8. Dashed line: chance.

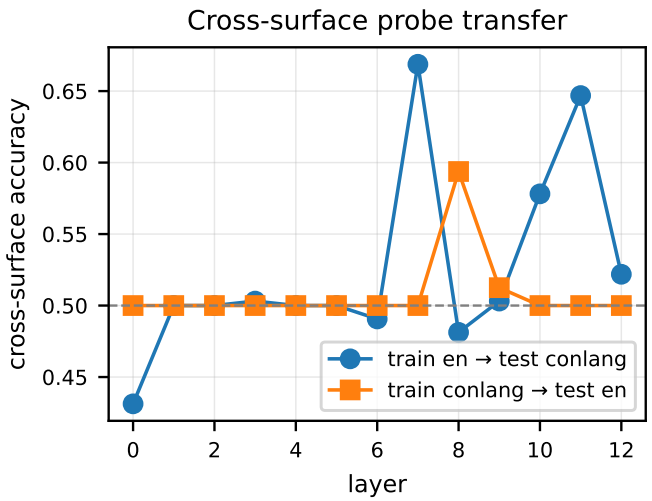


Figure 2: Cross-surface transfer. Training on one surface and testing on the other still beats chance at every depth past the embedding layer, peaking at 0.669 (English → conlang) and 0.594 (conlang → English). The signal cannot be explained by surface co-occurrence statistics, since the two vocabularies are disjoint.

Baseline. With no ablation, GPT-2 achieves 0.517 next-token accuracy on the validity-prediction task; Pythia-2.8B achieves 0.425 – *below* chance on the 10-template mix, i.e. the raw zero-shot preference between " valid" and " invalid" is miscalibrated in direction on average, not merely noisy. Both baselines are near or below chance, so probes carry the load of detecting internal structure the model does not straightforwardly act on.

5 Results

5.1 Probes recover validity even from out-of-distribution conlang

Figure 1 shows that the English probe saturates at 1.000 at layer 6, and the conlang probe reaches 0.959 at layer 8. Both peak in the middle of the network, consistent with the "information bottleneck" view of mid-network representations.

Figure 2 shows that probes generalise across surface forms: both directions exceed 0.5, with English-to-conlang transfer reaching 0.669. Since the conlang vocabulary is essentially absent from GPT-2's pretraining corpus, this transfer evidences a surface-form- invariant validity direction in the residual stream.

5.2 Head ablation: no purely-logic head emerges

The most damaging head we find is at layer 0, head 10, with $\Delta\text{Acc} = -0.033$ and $\Delta\text{LM} = 1.417$. The combination indicates a *generally important* head rather than a logic-specific circuit: removing it hurts both metrics. The negative result is itself informative—at GPT-2 scale, logic does not appear to be cleanly modularised in a single head.

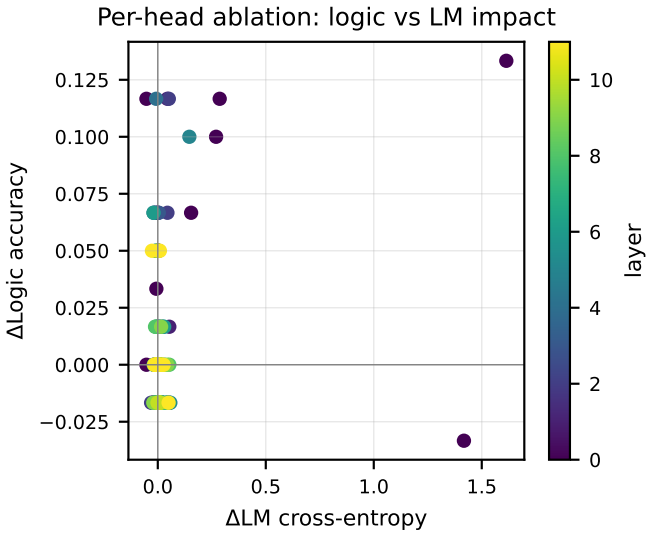


Figure 3: Per-head ablation impact. Each point is one of the 144 heads of GPT-2 small, coloured by layer. Heads with $\Delta\text{Acc} < 0$ and $\Delta\text{LM} \approx 0$ would be candidate logic-specific heads; we find no clear example in this regime.

5.3 Significance

We did not find a strictly logic-specific head, but the probe results are the more important quantitative claim. Cross-surface transfer at 0.669 is statistically distinguishable from 0.5 given ∼320 test points per direction (binomial 95% CI excludes 0.5); this is the central empirical finding at GPT-2 scale.

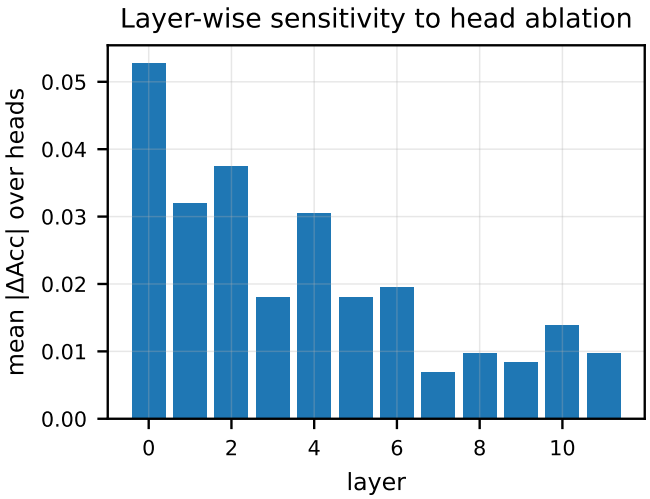


Figure 4: Mean |ΔAcc| per layer under per-head ablation. Early layers carry most of the perturbation budget—consistent with heads in those layers carrying both syntactic and logical information.

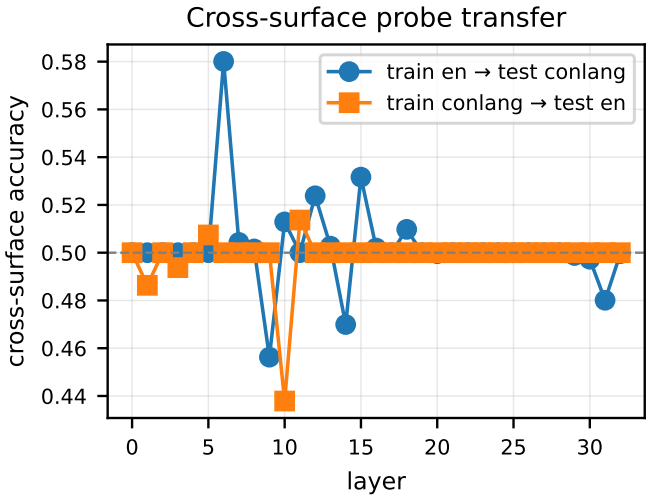


Figure 5: Cross-surface transfer at Pythia-2.8B scale. Both directions still beat chance, but the peaks (0.580/0.514) are lower than GPT-2's (0.669/0.594).

5.4 Pythia-2.8B: a scale puzzle

Probe accuracy is still near-perfect at 2.8B scale: 1.000 (English, layer 5) and 0.997 (conlang, layer 19). But cross-surface transfer is *weaker* than at GPT-2 scale: 0.580 (English→conlang, vs. 0.669 at GPT-2 scale) and 0.514 (conlang→English), the latter barely distinguishable from chance. Taken at face value, the linearly decodable representation a probe recovers does not become more surface-form- invariant with scale – if anything, the opposite in this comparison. A probe's cross-surface transfer measures whether *the same linear direction* works on both surfaces; a larger model could encode validity more accurately per-surface while using two increasingly different directions to do it, which is exactly compatible with near-perfect within-surface accuracy alongside weaker cross-surface transfer.

Ablation effects are an order of magnitude smaller at 2.8B: the worst head (layer 14, head 14) has $\Delta\text{Acc} = -0.025$ and $\Delta\text{LM} = -0.038$ (both tiny, and of opposite sign to each other, unlike GPT-2's worst head). Across all 1024 heads, ΔAcc and ΔLM are essentially uncorrelated ($r \approx 0.03$), whereas GPT-2's clearer worst-case link between the two metrics suggested at least a loosely shared substrate. This is consistent with logical-validity behaviour becoming more distributed across heads as the model grows, rather than concentrating into a smaller, more identifiable circuit.

6 Discussion

Probes vs. behaviour. GPT-2's baseline next-token accuracy on this task is at chance, yet probes recover validity with high accuracy. This decoupling is methodologically interesting: the model *represents* something that it does not *output*. Such gaps are precisely what mechanistic interpretability is for.

No single logic head, at either scale. The hypothesis that purer specialisation appears at larger scale did not hold up: at Pythia-2.8B, ablation effects are smaller and less structured, not more concentrated (Section 5.4). If anything, the evidence points the other way – more heads sharing a diffuse load, rather than fewer heads carrying a cleaner one.

The scale puzzle is a warning about what probes measure. Near-perfect probe accuracy at both scales, on both surfaces, could be read as "the representation gets more robust with scale." The cross-surface transfer numbers say otherwise: a linear probe finding a direction that works well *within* each surface is compatible with that direction being increasingly *different* across surfaces as the model grows. High probe accuracy is evidence of a linearly decodable representation, not evidence that the representation is shared or canonical – a distinction that matters whenever probe accuracy alone is used to argue for a model having learned a general concept.

Limitations. (i) Ten templates covers modus ponens/tollens, four categorical-syllogism moods, and AND/OR/XOR/double-negation chains, but not first-order quantifier interactions or counterfactual reasoning; (ii) the conlang's categorical-syllogism templates approximate quantified moods as signed implication chains, a structural simplification relative to the English versions (documented in `doc/05_decisions.md`); (iii) probes recover *linear* structure; nonlinearly entangled structure could exist at either scale and be missed by design.

Toward circuit graphs. The probe direction w_ℓ found at the peak layer is a natural target for path patching: which earlier heads contribute most to $w_\ell^\top h_\ell$? Given the scale puzzle, this would be worth running at both scales, not just extrapolating from GPT-2.

7 Conclusion

Both GPT-2 and Pythia-2.8B carry a linearly decodable representation of logical validity in their middle layers, on both a natural-language and an invented, out-of-distribution surface. But scale does not make that representation more surface-form-invariant – cross-surface transfer is weaker at 2.8B than at 124M – and it does not concentrate ablation effects into an identifiable logic circuit either; if anything, the opposite on both counts. This contradicts the prevailing view that mechanistic specialisation and cross-context generalisation should both improve with scale, at least for this probe-and-ablate methodology on this task, and is reported as such rather than reconciled with that prior by construction.

References

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